AIN SHAMS UNIVERSITY
FACULTY OF ENGINEERING

Mechatronics and Computer Programs
Spring 2026

CSE323: Advanced Embedded Systems Design
CSE411/CSE411s: Real-Time and Embedded Systems Design

Assignment-02

Description

You are required to prepare a **report with screenshots** demonstrating a **wrong context switching scenario on an ARM Cortex-M4 processor caused by incorrect Stack Pointer (SP) handling.**

The report must show how an **incorrect SP value during a context switch** leads to **state corruption or execution failure**.

What your report must demonstrate

- A Cortex-M4 execution scenario with **multiple tasks/functions**
- A **wrong context switch** caused by **incorrect SP restoration**
- **Incorrect execution behavior** resulting from the switch

  (e.g., popping wrong values from the stack, corrupted local variables, return address errors)
- **Supporting technical evidence** (debug view, or register/stack view)

Academic integrity notice

Submissions will be **checked for originality**.

Copied or reused submissions with highly similar content or narration, will be treated as **cheating** and handled according to faculty regulations.